```
daniel@DESKTOP-5SA89G3:~/arqui1$ make gdb
gdb-multiarch -q -x s.gdb programa
Reading symbols from programa...
The target architecture is set to "aarch64".

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Debuginfod has been disabled.
To make this setting permanent, add 'set debuginfod enabled off' to .gdbinit.
_start () at practica.s:386
386             ldr x1, =msg_inicial
Breakpoint 1 at 0x400538: file practica.s, line 483.
Breakpoint 2 at 0x4001ac: file leer_comando.s, line 24.
Breakpoint 3 at 0x4000c8: file interpretar.s, line 23.
Breakpoint 4 at 0x4003c4: file fork.s, line 17.
Breakpoint 5 at 0x4002d4: file execve.s, line 17.

=== Inicio: registros y stack ===
x0              0x0                 0
x1              0x0                 0
x2              0x0                 0
x8              0x0                 0
x30             0x0                 0
sp              0x7c69cfffee40      0x7c69cfffee40
0x7c69cfffee40: 0x0000000000000001      0x00007c69cffff0c9
0x7c69cfffee50: 0x0000000000000000      0x00007c69cffff0d4
0x7c69cfffee60: 0x00007c69cffff0fc      0x00007c69cffff104
0x7c69cfffee70: 0x00007c69cffff11c      0x00007c69cffff15d

Breakpoint 1, print () at practica.s:483
483             mov x0, #1              // stdout

=== Parada en print (prompt): x1 = mensaje, x2 = longitud ===
x0              0x0                 0
x1              0x4105f0            4261360
x2              0xf                 15
x8              0x0                 0
x30             0x400468            4195432
sp              0x7c69cfffee40      0x7c69cfffee40
0x7c69cfffee40: 0x0000000000000001      0x00007c69cffff0c9
0x7c69cfffee50: 0x0000000000000000      0x00007c69cffff0d4
0x7c69cfffee60: 0x00007c69cffff0fc      0x00007c69cffff104
0x7c69cfffee70: 0x00007c69cffff11c      0x00007c69cffff15d

Breakpoint 2, intrucion_init () at leer_comando.s:24
24          stp x29, x30, [sp, #-16]!
```

```
=== Parada en intrucion_init: x0 = bytes leidos ===
x0             0xa                 10
x9             0x0                 0
x16            0x410748            4261704
x17            0x410788            4261768
x30            0x400488            4195464
sp             0x7c69cfffee40      0x7c69cfffee40
0x7c69cfffee40: 0x0000000000000001      0x00007c69cffff0c9
0x7c69cfffee50: 0x0000000000000000      0x00007c69cffff0d4
0x7c69cfffee60: 0x00007c69cffff0fc      0x00007c69cffff104
0x7c69cfffee70: 0x00007c69cffff11c      0x00007c69cffff15d
26          mov x9, x0
0x7c69cfffee30: 0x0000000000000000      0x0000000000400488
0x7c69cfffee40: 0x0000000000000001      0x00007c69cffff0c9
0x7c69cfffee50: 0x0000000000000000      0x00007c69cffff0d4
0x7c69cfffee60: 0x00007c69cffff0fc      0x00007c69cffff104
(arriba: el stp ya guardo x29 y x30 en el stack)

Breakpoint 3, comparar () at interpretar.s:23
23          stp x29, x30, [sp, #-16]!

=== Parada en comparar: x16 = comando, x18 = longitud ===
x16            0x410748            4261704
x17            0x410788            4261768
x18            0x4                 4
x19            0x4                 4
x30            0x400490            4195472
sp             0x7c69cfffee40      0x7c69cfffee40
0x7c69cfffee40: 0x0000000000000001      0x00007c69cffff0c9
0x7c69cfffee50: 0x0000000000000000      0x00007c69cffff0d4
0x7c69cfffee60: 0x00007c69cffff0fc      0x00007c69cffff104
0x7c69cfffee70: 0x00007c69cffff11c      0x00007c69cffff15d
(gdb) continue
Continuing.

Breakpoint 1, print () at practica.s:483
483             mov x0, #1              // stdout
(gdb) continue
Continuing.

Breakpoint 1, print () at practica.s:483
483             mov x0, #1              // stdout
(gdb) continue
Continuing.

Breakpoint 1, print () at practica.s:483
483             mov x0, #1              // stdout
(gdb) continue
Continuing.

Breakpoint 2, intrucion_init () at leer_comando.s:24
24          stp x29, x30, [sp, #-16]!
(gdb) continue
Continuing.
```

```
Breakpoint 3, comparar () at interpretar.s:23
23          stp x29, x30, [sp, #-16]!
(gdb) continue
Continuing.

Breakpoint 4, fork () at fork.s:17
17          stp x29, x30, [sp, #-16]!
(gdb) continue
Continuing.
[Detaching after fork from child process 9815]

Breakpoint 1, print () at practica.s:483
483             mov x0, #1                  // stdout
(gdb) continue
Continuing.
Remote connection closed
(gdb) quit
daniel@DESKTOP-5SA89G3:~/arqui1$
```